Erratum: Equation 31 in "A Geometric Vocabulary for Gravitational and Dynamical Physics"

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Applies to: Nguyen, D. (2026). "A Geometric Vocabulary for Gravitational and Dynamical Physics: Circumferences, Geometric Means, and Spatial Dimensions Across 55 Equations." Zenodo.

Correction

Equation 31 of the original paper stated:

> **Original (Equation 31):** C(photon sphere) = (D/2) × C(event horizon), i.e., r_ph = (D/2) × r_s, where D = 3 is the number of spatial dimensions.

This decomposition is **incorrect for D ≠ 3** and should not be generalized to higher-dimensional gravity.

The correct higher-dimensional result comes from the Tangherlini metric (Tangherlini 1963), which generalizes the Schwarzschild solution to (D+1)-dimensional spacetime. In the Tangherlini solution, the photon sphere radius is:

$$r_{\text{ph}} = \left(\frac{D}{2}\right)^{1/(D-2)} \times r_s$$

not

$$r_{\text{ph}} = \frac{D}{2} \times r_s$$

These two expressions agree only at D = 3, where (3/2)^{1/(3-2)} = 3/2. For other values of D they diverge:

D	Original (D/2)	Correct (D/2)^{1/(D-2)}
3	1.500	1.500
4	2.000	1.414
5	2.500	1.357
6	3.000	1.316

The original paper's decomposition 6 = 2 × D (i.e., $r_{ISCO} = 6GM/c^2$ as $2 \times 3 \times GM/c^2$) and 3 = D ($r_{ph} = 3GM/c^2$ as $D \times GM/c^2$) are numerological coincidences specific to D = 3, not generalizable dimensional patterns.

Corrected classification

Equation 31 should be reclassified from **Speculative** to **Incorrect for D ≠ 3**. The same caution applies to Equation 12 (ISCO = D × horizon circumference), which also relies on Schwarzschild-specific factors that do not generalize to $(D/2) \times r_s$ in higher-dimensional gravity. In Kerr spacetime, r_{ISCO} depends on spin and ranges from 1 to 9 GM/c^2, as noted in the original paper.

The remaining 53 equations in the catalog are unaffected by this correction. In particular, the equations where D = 3 generalizes correctly to higher dimensions — Friedmann equation (#9), cosmological constant (#17), RMS speed (#53), Sackur-Tetrode entropy (#54) — are confirmed in the higher-dimensional physics literature and remain valid.

Reference

Tangherlini, F. R. (1963). Schwarzschild field in n dimensions and the dimensionality of space problem. *Nuovo Cimento*, 27, 636–651.

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